

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
 Section:

1. **Practice with 4-vectors.** (40 points) In lecture we defined the 4-vectors dx^μ , V^μ , A^μ , P^μ , and F^μ . Compute the following relativistic dot products among these 4-vectors, and verify that they are invariant by computing their value in at least two different reference frames. (Hint: a good choice for one of the reference frames is always the rest frame!)

(a) $V \cdot V$

In a general frame, $V^\mu = (\gamma, \gamma \vec{v})$. In the rest frame, $\vec{v} = 0$ and $\gamma = 1$, so $V^\mu = (1, \vec{0})$ and $V \cdot V = 1 - \vec{0} \cdot \vec{0} = 1$. In another frame with nonzero $\vec{v}$,

$$V \cdot V = \gamma^2 - \gamma^2 \vec{v} \cdot \vec{v} = \gamma^2(1 - v^2) = 1.$$

(b) $A \cdot A$ (Careful! The acceleration transforms between reference frames: see Morin Sec. 4.5.2.)

In the rest frame, we have $A^\mu = (0, \vec{a}')$, so $A \cdot A = -a'^2$. Note that $\vec{a}'$ is the acceleration experienced by a particle in its rest frame. Let's take the second frame to be one with velocity v_x along the x -direction only, and let $\vec{a} = (a_x, a_y, a_z)$ be the acceleration in this frame. Then $A^\mu = (\gamma^4 v_x a_x, \gamma^4 a_x, \gamma^2 a_y, \gamma^2 a_z)$ (Morin (4.9)) with $\gamma = 1/\sqrt{1 - v_x^2}$. To relate $\vec{a}$ to $\vec{a}'$, we use Morin (4.29): $a_x = a'_x/\gamma^3$, $a_y = a'_y/\gamma^2$, $a_z = a'_z/\gamma^2$. Thus in the frame with velocity v_x , we have $A^\mu = (\gamma v_x a'_x, \gamma a'_x, \gamma a'_y, \gamma a'_z)$. Taking the dot product,

$$\begin{aligned} A \cdot A &= \gamma^2 v_x^2 (a'_x)^2 - \gamma^2 (a'_x)^2 - (a'_y)^2 - (a'_z)^2 \\ &= (a'_x)^2 \gamma^2 (v_x^2 - 1) - (a'_y)^2 - (a'_z)^2 \\ &= -(a'_x)^2 - (a'_y)^2 - (a'_z)^2 \\ &= -a'^2 \end{aligned}$$

(c) $V \cdot F$ (The forces also transform between frames, see Morin Sec. 4.5.1.)

In the rest frame, $F^\mu = (0, \vec{F}')$ and $V^\mu = (1, \vec{0})$, so $V \cdot F = 0$. Let's again move to a frame where the velocity is v_x along the x -direction only, so $V^\mu = (\gamma, \gamma v_x, 0, 0)$. Transforming the forces using Morin (4.25) and using the general form $F^\mu = \gamma(dE/dt, \vec{F})$, we have $F^\mu = (\gamma v_x F'_x, \gamma F'_x, \gamma F'_y, \gamma F'_z)$. Taking the dot product,

$$V \cdot F = \gamma^2 v_x F'_x - \gamma^2 v_x F'_x = 0.$$

- (d) $F \cdot dx$, where dx represents an infinitesimal displacement along the world-line of the particle on which the force acts.

In the rest frame, $dx^\mu = (d\tau, \vec{0})$, so $F \cdot dx = 0$. Working in the v_x frame as in (b) and (c) above, we have $dx^\mu = (dt, v_x dt, 0, 0)$. A very similar calculation to (c) gives

$$F \cdot dx = \gamma v_x F'_x(dt) - \gamma F'_x(v_x dt) = 0.$$

2. **Relativistic snowballs.** (40 points) You are at the origin $(x, y) = (0, 0)$ in your frame S . Your friend, moving along the positive x -axis at velocity $\beta = 0.7$, throws a snowball of mass m at you at the instant he is located at $(x', y') = (-4, -1)$ in your frame, where distances are measured in light-seconds. In your friend's frame S' , the snowball is thrown with kinetic energy $3m$ at an angle θ with respect to the x -axis. In the following questions, give numerical values for your final answers, but you may want to keep your intermediate steps symbolic for clarity. Draw a diagram!

- (a) Find a quadratic equation that determines $\cos \theta$ in order for the snowball to hit you, and solve it to find the numerical value of $\cos \theta$.

Let's consider the snowball's velocity 4-vector V^μ . In frame S' , it has total energy $4m$ (kinetic energy $3m$ and rest energy m), so its gamma factor is $\gamma = 4$, and thus its speed is $v = \frac{\sqrt{\gamma^2 - 1}}{\gamma} = \frac{\sqrt{15}}{4}$. Its velocity 4-vector is then $V^\mu = (\gamma, \gamma \vec{v})$. Letting θ be the angle with which the snowball is thrown in S' , we have $V^\mu = (4, \sqrt{15} \cos \theta, \sqrt{15} \sin \theta, 0)$ (also in S'). Boosting this to your frame gives (letting $\gamma' = 1/\sqrt{1 - \beta^2} = 1.40$):

$$\Lambda^\mu_\nu V^\nu = \begin{pmatrix} \gamma' & \gamma'\beta & 0 & 0 \\ \gamma'\beta & \gamma' & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ \sqrt{15} \cos \theta \\ \sqrt{15} \sin \theta \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma'(4 + \beta\sqrt{15} \cos \theta) \\ \gamma'(4\beta + \sqrt{15} \cos \theta) \\ \sqrt{15} \sin \theta \\ 0 \end{pmatrix}.$$

To hit you, we want the ratio of the y to x components of the velocity in S to equal $y'/x' = 1/4$ (in other words, $\tan \theta_S = 1/4$ where θ_S is the angle in your frame). Thus

$$\frac{1}{4} = \frac{\sqrt{15} \sin \theta}{\gamma'(4\beta + \sqrt{15} \cos \theta)}.$$

Writing $u = \cos \theta$ and $\sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - u^2}$, this is a quadratic equation for $u = \cos \theta$. Plugging in numbers and rearranging, the equation is

$$269.4u^2 + 42.5u - 224.6 = 0$$

The negative root is spurious (if the snowball is thrown backwards it won't hit you), so we take the positive root $u = 0.838$. (Note that this implies that the angle in frame S' is about 33° , while the angle in your frame is $\theta_S \approx 14^\circ$: this is an example of relativistic beaming, where angles shrink in a boosted frame.)

- (b) How much time elapses in your frame before the snowball hits you?

In your frame S , the snowball has gamma factor γ_S equal to the zeroth component of its velocity 4-vector, which is $\gamma_S = \gamma'(4 + \beta\sqrt{15} \cos \theta) \approx 8.78$. Its speed in your frame is then

$$v_S = \sqrt{1 - \frac{1}{\gamma_S^2}} \approx 0.993.$$

The snowball is a distance $\ell = \sqrt{4^2 + 1^2} = 4.12$ light-seconds away from you when it is thrown, so the time elapsed is $\Delta t = \frac{\ell}{v_S} = 4.15$ seconds.

- (c) How much proper time elapses in the snowball's frame before the snowball hits you?

Using the invariant interval, $\Delta\tau = \sqrt{(\Delta t)^2 - \ell^2} = \ell\sqrt{\frac{1}{v_S^2} - 1}$ where v_S is given in part

(b). This is equivalent to $\Delta\tau = \frac{\ell/v_S}{\gamma_S} = \Delta t/\gamma_S \approx 0.47$ seconds, since this snowball's time is dilated by its gamma factor γ_S compared to your time.

- (d) What is the magnitude of the snowball's momentum in terms of m , in both frames S' and S ?

The energy-momentum 4-vector is just $P^\mu = mV^\mu$. In frame S' ,

$$P_{S'}^\mu = (4m, m\sqrt{15} \cos \theta, m\sqrt{15} \sin \theta, 0)$$

so the momentum is $\sqrt{15}m \approx 3.87m$. In frame S , we can use what we already calculated: the momentum is $\gamma_S m v_S \approx 8.73m$, which is greater than the momentum in S' . This is as expected, since the snowball receives a relativistic boost from the moving frame S' as it is thrown with an x -component of velocity in the same direction as the motion of S' .

3. Relativistic stoplight. (20 points)

- (a) You are in a spaceship on an intergalactic superhighway and you see a traffic light up ahead. You observe the light to be green (wavelength 550 nm) so you continue on your way. But you are pulled over by the space police, who tell you that you ran a red light! If the stoplight did indeed emit red light (wavelength 700 nm) in its rest frame, how fast were you going?

Using the longitudinal Doppler shift formula, we solve

$$\frac{550 \text{ nm}}{700 \text{ nm}} = \sqrt{\frac{1 + \beta}{1 - \beta}} \implies \beta = 0.237.$$

- (b) To show that this situation never arises in everyday life, calculate the smallest wavelength a human could have observed from a 700 nm red light, traveling relative to the light at the fastest speed humans have ever achieved, 40,000 km/hr (measured during the re-entry of Apollo 10 in 1969). Before you start, compare this speed to the speed of light, and consider whether a Taylor expansion of the Doppler formula would be appropriate! 40,000 km/hr is 1.1×10^4 m/s, or 3.7×10^{-5} in natural units. This is much less than 1, so we can Taylor-expand the Doppler formula to obtain the redshift expression from Discussion 10, $\lambda'/\lambda \approx 1 - \beta$. So

$$\lambda' \approx \lambda(1 - \beta) = 699.974 \text{ nm},$$

which is definitely still a red light!